

lec 16+2

Recall $i\hbar \frac{\partial \hat{U}(t)}{\partial t} = \hat{H} \frac{\partial \hat{U}}{\partial t}$

and $\hat{A}_H(t) = \hat{U}^\dagger(t) \hat{A}_S \hat{U}(t)$

$$\dot{\hat{A}}_H = \frac{[\hat{A}_H(t), \hat{H}_H(t)]}{i\hbar} + \left(\frac{\partial \hat{A}_S}{\partial t} \right)_H$$

↳ just another operator, so:

$$\dot{\hat{A}}_H = \hat{U}^\dagger \dot{\hat{A}}_S \hat{U} \Rightarrow \boxed{\frac{d\hat{A}_S}{dt} = \frac{[\hat{A}_S, \hat{H}_S]}{i\hbar} + \frac{\partial \hat{A}_S}{\partial t}}$$

Free particle $\hat{H} = \frac{\hat{p}^2}{2m} = \frac{\hat{p}_x^2 + \hat{p}_y^2 + \hat{p}_z^2}{2m}$

~~$\hat{p}_x$~~ $\dot{\hat{p}}_x = \frac{[\hat{p}_x, \hat{H}]}{i\hbar} = 0 \rightarrow \text{constant of motion.}$

~~$\hat{x}$~~ $\dot{\hat{x}} = \frac{[\hat{x}, \hat{H}]}{i\hbar} = -\frac{[\hat{p}_x^2, \hat{x}]}{i\hbar 2m} = +\frac{(\hat{x} \hat{p}_x \hat{p}_x - \hat{p}_x \hat{p}_x \hat{x})}{i\hbar 2m}$

$$= \frac{\hat{p}_x \hat{x} \hat{p}_x}{i\hbar 2m} + \frac{\hat{p}_x}{2m} - \frac{\hat{p}_x \hat{p}_x \hat{x}}{i\hbar 2m}$$

$$= \frac{\hat{p}_x}{2m} + \frac{\hat{p}_x}{2m i\hbar} [\hat{x}, \hat{p}_x] = \frac{\hat{p}_x}{m} \Rightarrow \boxed{\dot{\hat{x}} = \frac{\hat{p}_x}{m}} *$$

Equivalent to using $[x_i, F(\hat{p})] = i\hbar \frac{\partial F}{\partial p_i}$

$$[p_i, G(\hat{x})] = -i\hbar \frac{\partial G}{\partial x_i}$$

Solve $*$: $\boxed{\hat{x}(t) = \hat{x}(0) + \frac{\hat{p}_x(0)}{m} t}$

Even though $[x_i(0), x_j(0)] = 0$

we have $[\hat{x}_i(t), \hat{x}_i(0)] = \left[\frac{\hat{p}_i(0)t}{m}, x_i(0) \right] = -\frac{i\hbar t}{m}$

Recall

$[A, B] \rightarrow$ uncertainty relation:

$$\langle (\Delta A)^2 \rangle \langle (\Delta B)^2 \rangle \geq \frac{|\langle [A, B] \rangle|^2}{4}$$

$$\Rightarrow \langle (\Delta x_i)^2 \rangle_t \langle (\Delta x_i)^2 \rangle_0 \geq \frac{\hbar^2 t^2}{4m^2}$$

Add now $V(x)$: $\hat{H} = \frac{\hat{p}^2}{2m} + V(\vec{x})$

$$\frac{d\hat{p}_i}{dt} = \frac{[\hat{p}_i, V(x)]}{i\hbar} = -\frac{\partial V(\vec{x})}{\partial x_i}$$

$$\frac{d\hat{x}_i}{dt} = \frac{[\hat{x}_i, \frac{\hat{p}^2}{2m} + V(\vec{x})]}{i\hbar} = \frac{\hat{p}_i}{m}$$

$$\text{Again } \frac{d^2 \hat{x}_i}{dt^2} = \frac{d}{dt} \left(\frac{\hat{p}_i}{m} \right) = \frac{1}{m} \frac{d\hat{p}_i}{dt}$$

$$\Rightarrow m \frac{d^2 \hat{x}_i}{dt^2} = -\nabla V(\vec{x}) = F(\vec{x}) \quad \boxed{F = ma}$$

$$m \frac{d^2 \langle \hat{x} \rangle}{dt^2} = \frac{d \langle p \rangle}{dt} = -\langle \nabla V(\vec{x}) \rangle$$

Ehrenfest Theorem